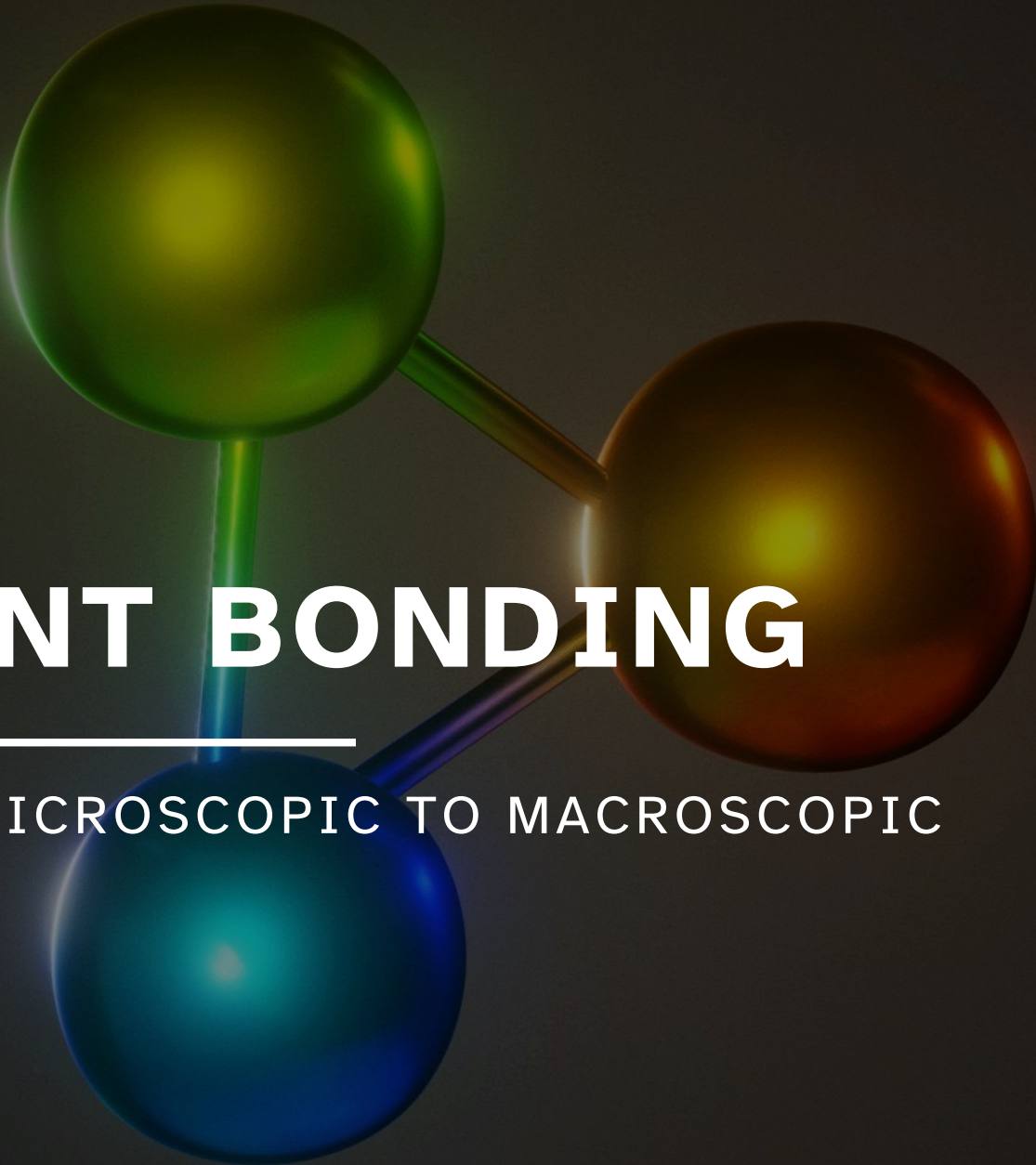




10A COVALENT BONDING

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 10

1. Predict the number of bonds an atom is likely to form in a molecule.
2. Draw Lewis structures for molecules that follow the octet rule.
3. Define and apply electronegativity and its periodic trends.
4. Draw Lewis structures for molecules for which multiple valid Lewis structures exist (called resonance).
5. Draw Lewis structures for molecules that do not obey the octet rule by application of the typical exceptions exhibited by H, Be, B, and expanded octets.
6. Determine bond polarity using electronegativity values.
7. Describe and apply the relation between bond length and bond strength.
8. Apply bond enthalpies to calculate enthalpies of reaction.

Where did we start and where are we going?

Chemistry in Context

What is our overall goal?



Formation of Covalent Bonds

10.1

Covalent bonds

- Formed by the *sharing of electrons* in the space between two atoms.

How is it possible for these electrons to count for both atoms involved in the covalent bond?

- Electrons are small (quantum) particles.
- Electron properties have *wave-like* probabilities.
- Electron position can be thought of as “smeared” 3D charge clouds called orbitals.
- Electrons are *shared* when the orbitals (or “smeared” electron charge clouds) overlap.

Formation of Covalent Bonds

10.1

Octet (8 e^-) rule

- Atoms with **eight valence electrons are particularly stable** because they have a full valence shell with an electron configuration that is:
- In a molecule, atoms will form covalent bonds so that each atom attains a *complement* of 8 electrons _____.

Duet (2 e^-) rule

- For H and He, the valence shell is $n = 1$ and can only hold two electrons.
- They can share *only* two electrons in molecules to achieve a $1s^2$ configuration.

Resonance and Electronegativity

10.3 (we are not covering formal charges)

Electronegativity (EN)

- The ability of an atom to attract *shared* electrons to itself in a bond.
- Different numerical scales exist, but Pauling scale is most common.
- Trends in EN values depend on many factors:

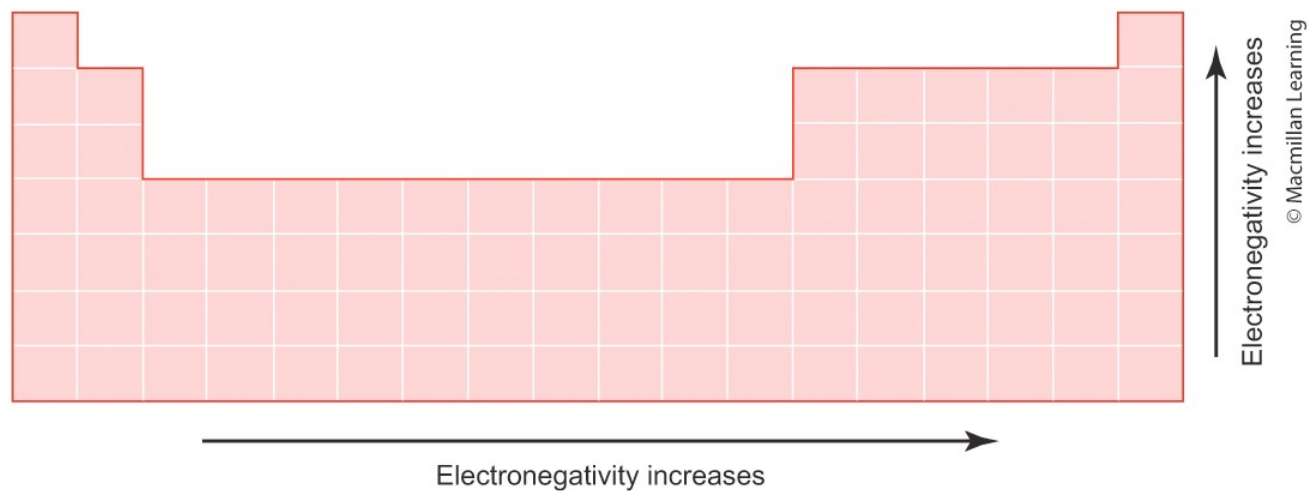


FIGURE 11.15 General electronegativity trend

Resonance and Electronegativity

10.3 (we are not covering formal charges)

An element with a *high electronegativity*:

- Pulls or attracts shared electrons closer to itself than an element with a lower electronegativity in a bond.

1 1A						18 8A
H 2.1	2 2A					He
Li 1.0	Be 1.5	13 3A	14 4A	15 5A	16 6A	17 7A
Na 0.9	Mg 1.2	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
K 0.8	Ca 1.0	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
Rb 0.8	Sr 1.0	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
Cs 0.7	Ba 0.9	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Fr 0.7	Ra 0.9	Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.2
						Xe 2.6
						Rn

Electronegativity increases

Electronegativity increases

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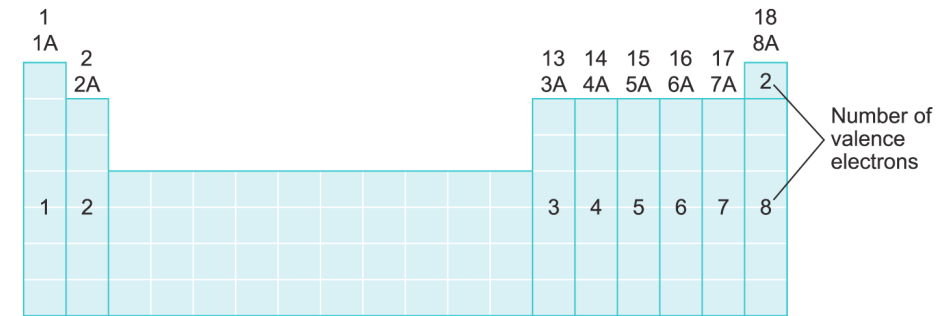
FIGURE 10.7 Pauling scale for electronegative values

Lewis Structures

10.2

Lewis structure: *localized electron* model for the structure of covalent molecules

- Only valence electrons
- Total number of e^- in Lewis structure *must equal* total number of valence e^- from all atoms
- Satisfy octet ($8 e^-$) and duet ($2 e^-$) rules



Atom	# of valence e^-	# of shared bonding e^-	# of unshared lone pair e^-	Total # of e^- around atom
Hydrogen H				
Fluorine F				

Formation of Covalent Bonds

10.1

Atoms can *share* different numbers of electrons to form different types of bonds.

Number of <i>shared</i> e^-	Bond Type	Representation	Example
	Single bond	—	H ₂ , F ₂
	Double bond	=	O ₂
	Triple bond	≡	N ₂

Lone pair: Set of 2 e^- that are *not shared* (not involved in bonding).

Lewis Structures

10.2

Method 1: Electron-Pushing

1. Count total number of valence e^- from all atoms.
2. Assign central atom and arrange all other atoms around it.
 - Central atom is *never* hydrogen.
 - Often: central atom is written *first* in the chemical formula, or it's the *least electronegative* atom.
3. Make skeleton: Use single bonds (—) to connect central atom to outer atoms.
4. Add lone pairs (·) to outer atoms to satisfy octet rule, or duet rule for hydrogen.
5. Satisfy octet rule for central atom.
 - a) If leftover e^- (from step 1), add as lone pairs (·) to central atom.
 - b) If *no* leftover e^- (from step 1), re-assign lone pairs from outer atoms to convert single bonds (—) to double bonds (=) or triple bonds ($\equiv$).
6. Verify Lewis structure is valid: count total number of e^- in structure and check octet/duet rule.

1 1A	2 2A												13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
1	2												3	4	5	6	7	8

Number of valence electrons

Lewis Structures

10.2

Method 2: Guess-and-Check

1. Count total number of valence e^- from all atoms.
2. Assign central atom and arrange all other atoms around it.
 - Central atom is *never* hydrogen.
 - Often: central atom is written *first* in the chemical formula, or it's the *least electronegative* atom.
3. Make skeleton: Use single bonds (—) to connect central atom to outer atoms.
4. Add lone pairs ($\cdot\cdot$) to outer atoms to satisfy octet rule, or duet rule for hydrogen.
5. Satisfy octet rule for central atom.
6. Count total number of e^- in structure drawn.
 - a) If correct number of electrons as step 1, then valid Lewis structure.
 - b) If *too many electrons* in structure, go back to step 3 and try double (=) and/or triple ($\equiv$) bonds.

Diagram illustrating the periodic table with the number of valence electrons indicated for each element. The groups are labeled 1A, 2A, 3A, 4A, 5A, 6A, 7A, and 8A. The number of valence electrons is shown in the corresponding cell for each element.

Group	1A	2A	Transition Metals (Groups 3-10)						3A	4A	5A	6A	7A	8A
Valence Electrons	1	2							3	4	5	6	7	8

Note: The number of valence electrons for elements in groups 3A through 7A is equal to the group number. For elements in group 8A, the number of valence electrons is 8.

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for H_2O .

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for NF_3 .

EXAMPLE PROBLEM

Learning Objective(s): 10.2, 10.4

Draw a valid Lewis structure for the carbonate anion (CO_3^{2-}).

IN-CLASS QUESTION 1

Learning Objective(s): 10.2, 10.3

Draw the Lewis structure for silicon tetrachloride (SiCl_4).

What is the total number of valence electrons in your Lewis structure?

What is the identity of the central atom?

IN-CLASS QUESTION 2

Learning Objective(s): 10.1, 10.2

Draw the Lewis structure for carbon monoxide (CO).

Complete the table below to describe the structure and bonding.

Atom		# of valence e^-	# of shared bonding e^-	# of unshared lone pair e^-	Total # of e^- around atom
Carbon	C				
Oxygen	O				

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Which elements could be X in the Lewis structure shown?

Select all that apply.

☐ C

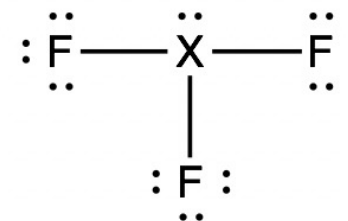
☐ N

☐ O

☐ F

☐ Si

☐ P



EXAMPLE PROBLEM

Learning Objective(s): 10.2, 10.3

Draw the Lewis structure for CHCl_2F .

Which statement is true regarding the atoms in this molecule?

- ☐ In the bond between carbon and hydrogen, hydrogen attracts electrons more strongly than carbon.
- ☐ In the bond between carbon and chlorine, carbon gets a larger share of the electron pair than chlorine.
- ☐ Fluorine has the greatest ability to attract electrons to itself in a bond.
- ☐ Carbon is the most electronegative atom.

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw the Lewis structure for C_2H_6 .

EXAMPLE PROBLEM

Learning Objective(s): 10.2

Draw two *different* Lewis structures for a molecule with the formula $\text{C}_2\text{H}_6\text{O}$.

BONUS QUESTION

Assessed on: Summer 2021, Exam 2

Draw the Lewis structures for CO_2 , H_2O , and O_2 .

- a) Which molecule has the greatest number of electrons in shared bonds?
- b) Which molecule has the smallest number of lone pair electrons?